

```
mysql -u'root' -p'password'
```

Your MySQL database installation and configuration is done.

Section 3 - Installing PHP-FPM and additional PHP extensions

We have Nginx which serves our Web pages and the MySQL database that stores and manages our data. But we still don't have anything to process dynamic Web pages. This is where PHP-FPM comes into play.

Ubuntu comes with one specific version of PHP, so it is a best practice to add a PHP repository. After adding the PHP repository, you can install any version of PHP. To add a repository and install the latest version of PHP with an extension, execute the following commands:

```
sudo apt-get install software-properties-common
sudo add-apt-repository ppa:ondrej/php
sudo apt-get install php7.4-fpm php7.4-mysql php7.4-curl php7.4-gd php7.4-intl php7.4-mbstring php7.4-soap php7.4-xml php7.4-xmlrpc php7.4-zip
```

Section 4 - Creating a MySQL database and user for WordPress

We now have all the things that run WordPress. In this section, let's create one database and user. Give privileges to that user on that database only. Execute the following command for this:

```
mysql -u'root' -p'password'

mysql> create database MyWordPress;
mysql> create user 'MyUser'@'localhost' IDENTIFIED BY 'password';
mysql> GRANT ALL PRIVILEGES ON MyWordPress.* TO 'MyUser'@'localhost';
mysql> FLUSH PRIVILEGES;
mysql> exit
```

You can check whether or not the user and database have been created

with the following command:

```
mysql -u'MyUser' -p'password'
```

```
mysql> show databases;
mysql> exit
```

Section 5 - Downloading WordPress and setting up the document directory for it

To download the WordPress content and set it in the document directory, execute the following commands:

```
cd /tmp
curl -O https://wordpress.org/latest.tar.gz
tar -xzf latest.tar.gz
cp /tmp/wordpress/wp-config-sample.php /tmp/wordpress/wp-config.php
mkdir /tmp/wordpress/wp-content/upgrade
mkdir /tmp/wordpress/wp-content/uploads
```

Create the directory in `/var/www/html`.

Best practice: Use a directory name that is the same as your domain or sub-domain that you want to deploy. Execute the following commands for this:

```
sudo mkdir /var/www/html/osfy-test.linuxgeek.in
sudo cp -a /tmp/wordpress/. /var/www/html/osfy-test.linuxgeek.in
```

Section 6 - Setting up the directory level permissions and ownership

Set ownership and file permissions for the document directory to make the site secure, so that the Nginx or PHP processes are able to access this content.

To do this, execute the following commands:

```
sudo chown -R your_user:www-data /var/www/html/osfy-test.linuxgeek.in
sudo find /var/www/html/osfy-test.linuxgeek.in -type d -exec chmod 750 {} \;
sudo find /var/www/html/osfy-test.
```

```
linuxgeek.in -type f -exec chmod 640 {} \;
```

Description of commands

- `chown`: Change owner and group owner of a file.
- `your_user`: This is the local user. You can get `your_user` by executing the `whoami` command.
- `www-data`: This is the group owner. It can be used by Web processes like Nginx and PHP-FPM.
- `chmod`: This command changes the permissions of the file.

Understanding the permissions in Octal

- Read - 4
- Write - 2
- Execute - 1
- No Permission - 0
- `find` command: Search for files in a directory hierarchy

Section 7 - Setting up the WordPress configuration file

Next, to configure WordPress, execute the following commands:

```
curl -s https://api.wordpress.org/secret-key/1.1/salt/
```

Copy the output of this command. After that, open the WordPress configuration file as follows:

```
vi /var/www/html/osfy-test.linuxgeek.in/wp-config.php
```

In that file, you will get similar lines, which you copy. Remove those lines and paste what you copied (output of the `curl` command).

Modify the database connection configuration in that file, as follows:

```
/** The name of the database for WordPress */
define( 'DB_NAME', MyWordPress );

/** MySQL database username */
define( 'DB_USER', 'MyUser' );
```